

PROJECT I—VANILLA CALL AND PUT OPTIONS FOR CRYPTOCURRENCIES

1 Objective

Price vanilla call and put options with Bitcoin / ETH as the underlying asset. Both European and American style options should be priced.

2 Data source

- Bitcoin / ETH price:
<https://finance.yahoo.com/quote/BTC-USD/history?p=BTC-USD>;
<https://finance.yahoo.com/quote/ETH-USD/history?p=ETH-USD>
- Options data: <https://www.deribit.com/options/BTC/BTC-12FEB26>

3 Questions

1. Download underlying asset's price data, analyze the data, estimate log return mean, volatility, and other parameter values. You may consider two risk-free rates $r = 1\%$ and $r = 5\%$ in your project.
2. Use binomial tree method (BTM) / Black-Scholes PDE to price European call and put options. Discuss numerically the convergence of the BTM price / Black-Scholes price and the computing-time for different number of periods, strike prices, Bitcoin/ETH prices, and time-to-maturities. Use Black-Scholes price for European call and put options as benchmark.
3. Download market option prices. Compare your BTM price with the market option price. Do your BTM prices / Black-Scholes prices match the market option prices well? Use proper measure to compute the difference between BTM prices / Black-Scholes price and market prices. Explain why you BTM prices / Black-Scholes price are different from market option prices?
4. Note that there are market prices for vanilla call and put options. Try to develop some trading strategies given these financial contracts. Report the performance of your strategy with historical data. For instance, you may consider use put-call parity to construct some trading strategy.
5. Use binomial tree method (BTM) to price American call and put options. Discuss numerically the convergence of the BTM price and the computing-time for different number of periods, strike prices, asset prices, and time-to-maturities.
6. Risk management analysis: compute the Greeks of vanilla options for both Bitcoin and ETH and compare the Greeks for Bitcoin and for ETH.
7. Numerically compute the implied volatility. Discuss the differences of implied volatility between options for Bitcoin and for ETH.
8. Bitcoin / ETH prices have many jumps, will hedging with vanilla options help to mitigate such risk?

4 Requirements on project report

1. Introduction: a summary of research objective, methodology, and findings
2. Methodology and data: the pricing method, data description, and estimation method used in your project
3. Project findings: data analysis, pricing results and analysis, and discussions.

4. Conclusion.